

2

Fractions

In order to understand arithmetic in general, it is important to practice and become comfortable with manipulating fractions. The problems in this chapter help you practice how to perform basic operations with fractions and will assist you in understanding real-world applications involving them.

- 64.** Lori ran $5\frac{1}{2}$ miles Monday, $6\frac{1}{4}$ miles Tuesday, $4\frac{1}{2}$ miles Wednesday, and $2\frac{3}{4}$ miles on Thursday. What is the average number of miles Lori ran?
- a. 5
 - b. $4\frac{1}{2}$
 - c. 4
 - d. $4\frac{3}{4}$
- 65.** Last year Jonathan was $59\frac{5}{8}$ inches tall. This year he is $64\frac{3}{8}$ inches tall. How many inches did he grow?
- a. $5\frac{1}{2}$
 - b. $4\frac{3}{4}$
 - c. $4\frac{1}{4}$
 - d. $5\frac{3}{4}$

- 66.** Larry spends $\frac{3}{4}$ hour twice a day walking and playing with his dog. He also spends $\frac{1}{6}$ hour twice a day feeding his dog. How much time does Larry spend on activities involving his dog each day?
- $\frac{11}{12}$ hour
 - $1\frac{1}{2}$ hours
 - $1\frac{5}{6}$ hours
 - $1\frac{2}{5}$ hours
- 67.** The first section of a newspaper is comprised of 16 pages. Advertisements take up $3\frac{3}{8}$ of the pages. How many pages are not advertisements?
- $12\frac{5}{8}$
 - $19\frac{3}{8}$
 - 13
 - $12\frac{1}{2}$
- 68.** Lisa was assigned 48 pages to read for English class. She has finished $\frac{3}{4}$ of the assignment. How many more pages must she read?
- 36
 - 21
 - 12
 - 8
- 69.** Mark has three $4\frac{1}{2}$ oz cans of tomatoes and five $8\frac{1}{4}$ oz cans. How many ounces of tomatoes does Mark have?
- $12\frac{3}{4}$
 - $54\frac{3}{4}$
 - 54
 - $62\frac{1}{4}$
- 70.** Joe walked $2\frac{3}{4}$ miles to school, $1\frac{1}{3}$ miles to the library, and $1\frac{2}{5}$ miles to his friend's house. How many miles did Joe walk altogether?
- $4\frac{1}{2}$
 - $5\frac{29}{60}$
 - $5\frac{3}{4}$
 - $4\frac{3}{5}$

- 71.** Justin read $\frac{1}{8}$ of a book the first day, $\frac{1}{3}$ the second day, and $\frac{1}{4}$ the third day. On the fourth day he finished the book. What fraction of the book did Justin read on the fourth day?
- a. $\frac{2}{5}$
 - b. $\frac{3}{8}$
 - c. $\frac{7}{24}$
 - d. $\frac{17}{24}$
- 72.** Tammi babysat for $5\frac{1}{2}$ hours. She charged \$7 an hour. How much should she get paid?
- a. \$35.50
 - b. \$42
 - c. \$35
 - d. \$38.50
- 73.** Tom needs to spread grass seed across his $3\frac{3}{4}$ acres of property. If he uses $2\frac{2}{3}$ bags of grass seed per acre, how many bags of grass seed does Tom need to purchase?
- a. 7
 - b. 8
 - c. 9
 - d. 10
- 74.** One batch of scones will provide enough for $\frac{3}{11}$ of Beth's Sunday brunch gathering of 44 people. If she bakes five batches of scones, how many additional people could she feed?
- a. 16
 - b. 11
 - c. 8
 - d. 60

- 75.** Allison baked a cake for Guy's birthday; $\frac{4}{7}$ of the cake was eaten at the birthday party. The next day, Guy ate one third of what was left. How much of the original cake did Guy eat the next day?
- a. $\frac{1}{7}$
 - b. $\frac{3}{7}$
 - c. $\frac{1}{4}$
 - d. $\frac{4}{21}$
- 76.** Josh practiced his clarinet for $\frac{5}{6}$ of an hour. How many minutes did he practice?
- a. 83
 - b. 50
 - c. 8.3
 - d. 55
- 77.** Tike Television has six minutes of advertising space every 15 minutes. How many $\frac{3}{4}$ -minute commercials can be fit into the six-minute block?
- a. $4\frac{1}{2}$
 - b. 8
 - c. 20
 - d. 11
- 78.** Betty grew $\frac{3}{4}$ inch over the summer. Her friends also measured themselves and found that Susan grew $\frac{2}{5}$ inch, Mike grew $\frac{5}{8}$ inch, and John grew $\frac{1}{2}$ inch. List the friends in order of who grew the least to who grew the most.
- a. Betty, John, Mike, Susan
 - b. Susan, Mike, John, Betty
 - c. John, Mike, Betty, Susan
 - d. Susan, John, Mike, Betty

- 79.** An industrial-sized box of cereal contains 21 cups of cereal. If a serving size is $\frac{3}{4}$ -cup, how many servings does the box contain?
- a. Nearly 16
 - b. 28
 - c. 21
 - d. 14
- 80.** The Grecos are taking an $8\frac{3}{4}$ mile walk. If they walk at an average speed of $3\frac{1}{2}$ miles per hour, how long will it take them?
- a. $2\frac{2}{3}$ hours
 - b. $30\frac{1}{8}$ hours
 - c. $2\frac{1}{2}$ hours
 - d. 5 hours
- 81.** The town's annual budget totals \$32 million. If $\frac{2}{5}$ of the budget goes toward education, how many dollars go to education?
- a. $\$6\frac{1}{2}$ million
 - b. $\$9\frac{1}{5}$ million
 - c. \$16 million
 - d. $\$12\frac{4}{5}$ million
- 82.** A recipe that makes 4 dozen snickerdoodles requires $2\frac{1}{3}$ cups of flour. How many cups of flour would be needed to bake $2\frac{1}{2}$ times the recipe?
- a. $9\frac{1}{3}$
 - b. $4\frac{2}{3}$
 - c. $23\frac{1}{3}$
 - d. $5\frac{5}{6}$
- 83.** Linus wants to buy ribbon to make three bookmarks. One bookmark will be $11\frac{1}{2}$ inches long and the other two will be $8\frac{1}{4}$ inches long. How much ribbon should he buy?
- a. 27 inches
 - b. 19 inches
 - c. $19\frac{3}{4}$ inches
 - d. 28 inches

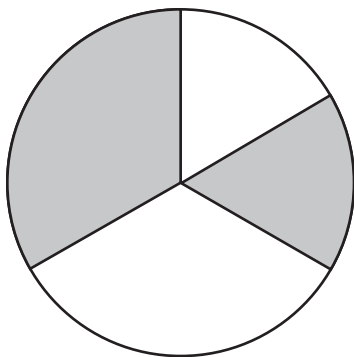
- 84.** Rita caught fish that weighed $3\frac{1}{4}$ lb, $8\frac{1}{2}$ lb, and $4\frac{2}{3}$ lb. What was the total weight of all the fish that Rita caught?
- $15\frac{4}{9}$ lb
 - $14\frac{2}{3}$ lb
 - $16\frac{5}{12}$ lb
 - 15 lb
- 85.** The dimensions of a rectangular garden are $4\frac{1}{2}$ yards by 3 yards. How many yards of fence are needed to surround the garden?
- $16\frac{1}{2}$
 - $7\frac{1}{2}$
 - 15
 - 14
- 86.** The soccer team is making pizzas for a fundraiser. They put $\frac{1}{4}$ of a package of cheese on each pizza. If they have 12 packages of cheese, how many pizzas can they make?
- 48
 - 3
 - 24
 - $11\frac{3}{4}$
- 87.** Dan purchased $3\frac{1}{2}$ yards of mulch for his garden. Mulch costs \$25 a yard. How much did Dan pay for his mulch?
- \$75.00
 - \$87.50
 - \$64.25
 - \$81.60
- 88.** Ten tablespoons of instant iced tea mix are needed to make one pitcher of iced tea. One pitcher of iced tea can fill 6 glasses. How many tablespoons of instant iced tea mix are needed to fill $1\frac{1}{2}$ glasses?
- $1\frac{1}{9}$
 - $2\frac{1}{2}$
 - 3
 - $3\frac{1}{4}$

- 89.** Lucy worked $32\frac{1}{2}$ hours last week and earned \$195. What is her hourly wage?
- a. \$7.35
 - b. \$5.00
 - c. \$6.09
 - d. \$6.00
- 90.** A sheet of plywood is $5\frac{1}{2}$ feet wide and $7\frac{1}{2}$ feet long. What is the area of the sheet of plywood?
- a. $41\frac{1}{4}$ sq ft
 - b. $82\frac{1}{2}$ sq ft
 - c. $35\frac{1}{4}$ sq ft
 - d. 11 sq ft
- 91.** Rosa kept track of how many hours she spent reading during the month of August. The first week she read for $4\frac{1}{2}$ hours, the second week for $3\frac{3}{4}$ hours, the third week for $8\frac{1}{5}$ hours, and the fourth week for $1\frac{1}{3}$ hours. How many hours altogether did she spend reading in the month of August?
- a. $17\frac{47}{60}$
 - b. 16
 - c. $16\frac{1}{8}$
 - d. $18\frac{2}{15}$
- 92.** During a commercial break in the Super Bowl, there were four $\frac{1}{2}$ -minute commercials and three $\frac{1}{4}$ -minute commercials. How many minutes was the commercial break?
- a. $2\frac{3}{4}$
 - b. $\frac{3}{4}$
 - c. $3\frac{1}{2}$
 - d. 5

- 93.** Refinancing a home loan amounted to reducing the monthly payment by $\frac{1}{9}$ of the present monthly payment. If the new monthly payment is \$1,128, how much was the original monthly payment?
- a. \$1,340
 - b. \$1,213
 - c. \$1,269
 - d. \$1,225
- 94.** Michelle is making a triple batch of chocolate chip cookies. The original recipe calls for $\frac{3}{4}$ cup of brown sugar. How many cups should she use if she is tripling the recipe?
- a. $\frac{1}{4}$
 - b. 3
 - c. $2\frac{1}{4}$
 - d. $1\frac{3}{4}$
- 95.** The output capacity of the equivalent of $1\frac{1}{3}$ printers is 15 pages per minute. How many pages would six printers, working together, produce in 8 minutes?
- a. 120
 - b. 480
 - c. 540
 - d. 800
- 96.** The Cheshire Senior Center is hosting a bingo night; \$2,400 in prize money will be given away. The first winner of the night will receive $\frac{1}{3}$ of the money. The next ten winners will receive $\frac{1}{10}$ of the remaining amount. How many dollars will each of these ten winners receive?
- a. \$240
 - b. \$800
 - c. \$160
 - d. \$200

- 97.** $\frac{3}{8}$ of the employees in the Acme Insurance Company work in the accounting department. What fraction of the employees does NOT work in the accounting department?
- a. $\frac{10}{16}$
 - b. $\frac{5}{16}$
 - c. $\frac{3}{4}$
 - d. $\frac{1}{2}$
- 98.** A bag contains 44 cookies. Kyle eats 8 of the cookies. What fraction of the cookies is left?
- a. $\frac{5}{6}$
 - b. $\frac{9}{11}$
 - c. $\frac{8}{9}$
 - d. $\frac{3}{4}$
- 99.** Marci's car has a 12-gallon gas tank. Her fuel gauge says that there is $\frac{1}{8}$ of a tank left. How many gallons of gas are left in the tank?
- a. 3
 - b. $1\frac{2}{3}$
 - c. $\frac{3}{4}$
 - d. $1\frac{1}{2}$
- 100.** Joey, Aaron, Barbara, and Stu have been collecting pennies and putting them in identical containers. Joey's container is $\frac{3}{4}$ full, Aaron's is $\frac{3}{5}$ full, Barbara's is $\frac{2}{3}$ full, and Stu's is $\frac{2}{5}$ full. Whose container has the most pennies?
- a. Joey
 - b. Aaron
 - c. Barbara
 - d. Stu

- 101.** What fraction of the circle below is shaded?



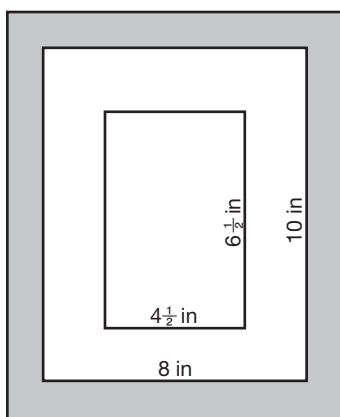
- a. $\frac{2}{3}$
b. $\frac{1}{2}$
c. $\frac{5}{6}$
d. $\frac{1}{3}$
- 102.** Michelle's brownie recipe calls for $2\frac{3}{4}$ cup of sugar. How much sugar does she need if she triples the recipe?
- a. $6\frac{1}{4}$ cups
b. $7\frac{3}{4}$ cups
c. $5\frac{1}{2}$ cups
d. $8\frac{1}{4}$ cups
- 103.** The output capacity of the equivalent of $1\frac{1}{4}$ printers is 20 pages per minute. The equivalent of how many printers would need to work together in order to produce 1,212 pages in three minutes?
- a. $25\frac{1}{4}$
b. $15\frac{3}{4}$
c. $12\frac{1}{6}$
d. $22\frac{1}{2}$

- 104.** Mr. Reynolds owns $1\frac{3}{4}$ acres of land. He plans to buy the property next to his, which is $2\frac{1}{8}$ acres. How many acres will Mr. Reynolds own after the purchase?
- a. $5\frac{1}{4}$
 - b. $3\frac{3}{4}$
 - c. $3\frac{1}{3}$
 - d. $3\frac{7}{8}$
- 105.** If Allen gets three oil changes in an eight-month period, how many oil changes will he have gotten at the end of $4\frac{2}{3}$ years?
- a. 7
 - b. 14
 - c. 21
 - d. 28
- 106.** How many eighths are in $4\frac{5}{8}$?
- a. 45
 - b. 48
 - c. 37
 - d. 9
- 107.** Kim is baking cookies for a large party and wants to double the recipe. The original recipe calls for $\frac{2}{3}$ cup of margarine. How many cups should she use for the double batch?
- a. $\frac{5}{6}$
 - b. $1\frac{2}{3}$
 - c. $1\frac{1}{3}$
 - d. $\frac{4}{6}$
- 108.** Mr. Johnson owns $4\frac{3}{4}$ acres. He sells one-third of his land. How many acres does he own after the sale?
- a. $2\frac{1}{4}$
 - b. $2\frac{3}{16}$
 - c. $3\frac{2}{3}$
 - d. $3\frac{1}{6}$

- 109.** Tim, Bob, and Fred return home for a short visit and, in anticipation, their mother bakes 12 dozen cookies. Tim arrived home first and ate $\frac{1}{9}$ of the entire batch of cookies. Bob arrived next and gathered up $\frac{3}{16}$ of what remained and took them to a party. Fred was the last to arrive home and took $\frac{4}{13}$ of what remained back to school with him. What fraction of cookies remained after Tim, Bob, and Fred had all taken their share?
- a. $\frac{7}{26}$
 - b. $\frac{1}{2}$
 - c. $\frac{2}{5}$
 - d. $\frac{5}{16}$
- 110.** The local firefighters are doing a “fill the boot” fundraiser. Their goal is to raise \$3,500. After 3 hours, they have raised \$2,275. Which statement below is accurate?
- a. They have raised 35% of their goal.
 - b. They have $\frac{7}{20}$ of their goal left to raise.
 - c. They have raised less than $\frac{1}{2}$ of their goal.
 - d. They have raised more than $\frac{3}{4}$ of their goal.
- 111.** Lori has half a pizza left over from dinner last night. For breakfast, she eats $\frac{1}{3}$ of the leftover pizza. What fraction of the original pizza remains after Lori eats breakfast?
- a. $\frac{1}{4}$
 - b. $\frac{1}{6}$
 - c. $\frac{1}{3}$
 - d. $\frac{3}{8}$
- 112.** A ball is dropped from a height of 108 feet. Each time it bounces, it rebounds to a height that is two-thirds of its previous height. What is the height of the ball after its fourth bounce?
- a. $1\frac{1}{3}$ feet
 - b. 64 feet
 - c. $10\frac{2}{3}$ feet
 - d. $21\frac{1}{3}$ feet

- 113.** A large coffee pot holds 120 cups. It is now about three-fifths full. About how many cups are currently in the pot?
- a. 24 cups
 - b. 72 cups
 - c. 48 cups
 - d. 90 cups
- 114.** An airport is backlogged with planes trying to land and has ordered planes to circle until they are told to land. A plane is using fuel at the rate of $9\frac{1}{2}$ gallons per hour, and it has $6\frac{1}{3}$ gallons left in its tank. How long can the plane continue to fly?
- a. $1\frac{1}{2}$ hours
 - b. $\frac{2}{3}$ hours
 - c. $2\frac{3}{4}$ hours
 - d. $\frac{3}{4}$ hours
- 115.** A carpenter receives measurements from a homeowner for a remodeling project. The homeowner lists the length of a room as $12\frac{3}{4}$ feet, but the carpenter would prefer to work in feet and inches. What is the measurement in feet and inches?
- a. 12 feet 9 inches
 - b. 12 feet 8 inches
 - c. 12 feet 6 inches
 - d. 12 feet 3 inches
- 116.** The state of Connecticut will pay two-thirds of the cost of a new school building. If the city of New Haven is building a school that will cost a total of \$15,300,000, how much will the state pay?
- a. \$7,750,000
 - b. \$5,100,000
 - c. \$10,200,000
 - d. \$1,020,000

- 117.** One-fourth of an inch on a map represents 150 miles. The distance on the map from Springfield to Oakwood is $3\frac{1}{2}$ inches. How many miles is it from Springfield to Oakwood?
- a. 600 miles
 - b. 2,100 miles
 - c. 1,050 miles
 - d. 5,250 miles
- 118.** Robert brings a painting to the framing store to be framed. He chooses a frame with a 8 in by 10 in opening. The painting is $4\frac{1}{2}$ in by $6\frac{1}{2}$ in. A mat will be placed around the painting to fill the 8 in by 10 in opening. If the painting is perfectly centered, what will the width of the mat be on each side of the painting? See the diagram below.



- a. $3\frac{1}{2}$ in
 - b. $2\frac{1}{4}$ in
 - c. $1\frac{1}{2}$ in
 - d. $1\frac{3}{4}$ in
- 119.** Mrs. Johnston's class broke into teams of three students each to play a game. The winning team received a $\frac{1}{2}$ -pound jar of jellybeans. How many pounds of jellybeans will each team member get if the $\frac{1}{2}$ -pound jar is shared equally among the three teammates?
- a. $\frac{1}{3}$ pound
 - b. $\frac{1}{6}$ pound
 - c. $\frac{2}{5}$ pound
 - d. $\frac{2}{9}$ pound

- 120.** Chuck is constructing a patio using $1\frac{1}{2}$ foot cement squares. The patio will be 10 cement squares by 10 cement squares. If the cement squares are placed right next to each other without any space in between, what will the dimensions of the patio be?
- a. 10 ft by 10 ft
 - b. 20 ft by 20 ft
 - c. $12\frac{1}{2}$ ft by $12\frac{1}{2}$ ft
 - d. 15 ft by 15 ft
- 121.** Samantha's Girl Scout troop is selling holiday wreaths. Each wreath has a bow that uses $\frac{2}{5}$ yard of ribbon. How many bows can Samantha make from a spool of ribbon that is 10 yards long?
- a. 25
 - b. 4
 - c. 20
 - d. 8
- 122.** Lindsay and Mark purchased a $\frac{3}{4}$ -acre plot of land to build a house. Zoning laws require that houses built on less than 1 acre take up no more than half the land. What is the largest amount of land that Lindsay and Mark's house can cover?
- a. $\frac{3}{8}$ acre
 - b. $\frac{1}{2}$ acre
 - c. $\frac{1}{4}$ acre
 - d. $\frac{5}{8}$ acre
- 123.** Mr. Grove was watching the price of a stock he recently bought. On Monday, the stock was at $26\frac{3}{8}$. By Friday, the stock had fallen to $23\frac{1}{16}$. How much did the stock price decline?
- a. $2\frac{5}{8}$
 - b. $3\frac{5}{16}$
 - c. $1\frac{3}{16}$
 - d. $3\frac{1}{16}$

- 124.** Four classmates are lined up in a row, and the first child in line is given a full 8 ounces of orange juice. He is instructed to pour $\frac{3}{4}$ of his glass into the glass of the next child in line. In turn, that child was instructed to pour $\frac{1}{2}$ of her glass into the glass of the next child in line. Finally, in turn, that child was instructed to pour $\frac{2}{3}$ of his glass's contents into the last child's glass. How many ounces of orange juice does the fourth child have in his glass?
- a. 1 ounce
 - b. 2 ounces
 - c. 3 ounces
 - d. 4 ounces
- 125.** Land in a development is setting for \$60,000 per acre. If Jack purchases $1\frac{3}{4}$ acres, how much will he pay?
- a. \$45,000
 - b. \$135,000
 - c. \$105,000
 - d. \$125,000

Answer Explanations

- 64. d.** To find the average, add the miles Lori ran each day and divide by the number of days. To add the fractions, use a common denominator of 4: $5\frac{2}{4} + 6\frac{1}{4} + 4\frac{2}{4} + 2\frac{3}{4} = 17\frac{8}{4} = 17 + 2 = 19$. Divide the sum by 4: $19 \div 4 = 4\frac{3}{4}$.
- 65. b.** Subtract to find the difference in heights. You will need to borrow 1 whole $\frac{8}{8}$ from 64 and add it to $\frac{3}{8}$ to make the fractional part of the mixed number $\frac{11}{8}$.
 $64\frac{3}{8} - 59\frac{5}{8} = 63\frac{11}{8} - 59\frac{5}{8} = 4\frac{6}{8} = 4\frac{3}{4}$
 If you chose **d**, you did not borrow, but rather simply subtracted the smaller fraction from the larger fraction.
- 66. c.** Add the times together to find the total amount of time. Remember that he walks the dog twice and feeds the dog twice. The common denominator is 12.
 $\frac{3}{4} + \frac{3}{4} + \frac{1}{6} + \frac{1}{6} = \frac{9}{12} + \frac{9}{12} + \frac{2}{12} + \frac{2}{12} = \frac{22}{12} = 1\frac{10}{12} = 1\frac{5}{6}$
 If you chose **a**, you did not consider that he walks and feeds the dog *twice* a day.
- 67. a.** Subtract the number of pages of advertisements from the total number of pages. Use a common denominator of 8 and borrow one whole ($\frac{8}{8}$) from 16 to do the subtraction.
 $16 - 3\frac{3}{8} = 15\frac{8}{8} - 3\frac{3}{8} = 12\frac{5}{8}$
- 68. c.** If Lisa has read $\frac{3}{4}$ of the assignment, she has $\frac{1}{4}$ left to go. To find $\frac{1}{4}$ of a number, divide the number by 4; $48 \div 4 = 12$ pages. If you chose **a**, you found the number of pages that she already read.
- 69. b.** Ignore the fractional parts of the mixed numbers at first and multiply the whole number portion of the ounces by the corresponding number of cans: $4 \times 3 = 12$ ounces and $8 \times 5 = 40$ ounces. Adding together 12 and 40, gives a total of 52 ounces. Next, find the fractional portion. Multiply the fractional part by the corresponding number of cans: $\frac{1}{2} \times 3 = \frac{3}{2} = 1\frac{1}{2}$ ounces and $\frac{1}{4} \times 5 = \frac{5}{4} = 1\frac{1}{4}$ ounces. Add together these

fractional parts: $1\frac{1}{2} + 1\frac{1}{4} = 2\frac{3}{4}$. Add this answer to the answer from the whole numbers to get the final answer: $2\frac{3}{4} + 52 = 54\frac{3}{4}$ ounces. If you chose **a**, you did not consider that Mark has THREE of the smaller cans and FIVE of the larger cans.

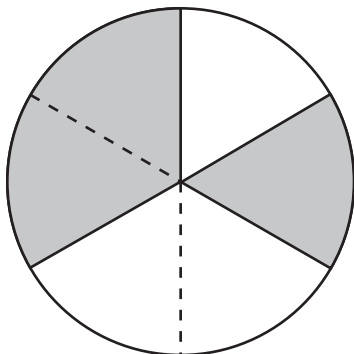
- 70. b.** To find the total distance Joe walked, add the three distances together using a common denominator of 60: $2\frac{45}{60} + 1\frac{20}{60} + 1\frac{24}{60} = 4\frac{89}{60}$, which is then simplified to $5\frac{29}{60}$.
- 71. c.** First, find the fraction of the book that Justin read the first 3 days by adding the three fractions using a common denominator of 24: $\frac{3}{24} + \frac{8}{24} + \frac{6}{24} = \frac{17}{24}$. Subtract this fraction of the book from one whole, using a common denominator of 24: $\frac{24}{24} - \frac{17}{24} = \frac{7}{24}$. If you chose **d**, you found the fraction of the book that Justin read the first 3 days.
- 72. d.** Multiply the hours Tammi babysat by the charge per hour. Change the mixed number to an improper fraction before multiplying: $\frac{11}{2} \times \frac{7}{1} = \frac{77}{2}$, which simplifies to $38\frac{1}{2}$ or \$38.50.
- 73. d.** Multiply the number of bags needed for one acre by the number of acres Tom has:
 $3\frac{3}{4} \times 2\frac{2}{3} = \frac{15}{4} \times \frac{8}{3} = \frac{120}{12} = 10$ bags.
- 74. a.** First, determine how many people corresponds to $\frac{3}{11}$ of the gathering by multiplying $\frac{3}{11}$ by 44: $\frac{3}{11} \times 44 = 12$ people. Next, let x stand for the number of people Beth can feed using 5 batches of scones. Set up the proportion $\frac{\text{batches}}{\text{people}}$ to solve for x : $\frac{1}{12} = \frac{5}{x}$. Cross multiply to solve for x : $60 = x$. So, five batches of scones will feed 60 people. Finally, to determine the number of people *beyond* the 44 attending the brunch gathering that Beth could feed, subtract 60 and 44: $60 - 44 = 16$ additional people.
- 75. a.** Find the uneaten part of the cake by subtracting the eaten part from one whole: $1 - \frac{4}{7} = \frac{7}{7} - \frac{4}{7} = \frac{3}{7}$ of the cake was uneaten. To find one third of this amount, multiply by $\frac{1}{3}$: $\frac{3}{7} \times \frac{1}{3} = \frac{1}{7}$.
- 76. b.** An hour is 60 minutes. To find the number of minutes in $\frac{5}{6}$ of an hour, multiply 60 by $\frac{5}{6}$: $\frac{60}{1} \times \frac{5}{6} = \frac{300}{6} = 50$ minutes.

- 77. b.** Divide the 6-minute block by $\frac{3}{4}$, remembering to take the reciprocal of the second fraction and multiply: $6 \div \frac{3}{4} = \frac{6}{1} \times \frac{4}{3} = \frac{24}{3} = 8$.
- 78. d.** To compare the fractions, use the common denominator of 40. Therefore, Betty = $\frac{30}{40}$, Susan = $\frac{16}{40}$, Mike = $\frac{25}{40}$, and John = $\frac{20}{40}$. To order the fractions, compare their numerators.
- 79. b.** To find the number of servings, divide the number of cups of cereal contained within the box (21) by the fraction of a cup equivalent to a single serving ($\frac{3}{4}$): $21 \div \frac{3}{4} = 21 \times \frac{4}{3} = \frac{84}{3} = 28$ servings.
- 80. c.** To find the amount of time that it took the Grecos to complete their walk, divide the distance ($8\frac{3}{4}$ miles) by the rate ($3\frac{1}{2}$ miles per hour). To divide mixed numbers, change them into improper fractions, then take the reciprocal of the second fraction and multiply:
- $$8\frac{3}{4} \div 3\frac{1}{2} = \frac{35}{4} \div \frac{7}{2} = \frac{35}{4} \times \frac{2}{7} = \frac{5}{2} = 2\frac{1}{2} \text{ hours.}$$
- 81. d.** To find $\frac{2}{5}$ of \$32 million, multiply the two numbers: $\frac{2}{5} \times \frac{32}{1} = \frac{64}{5}$: which simplifies to \$12 $\frac{4}{5}$ million.
- 82. d.** Multiply the number of cups of flour needed for a single recipe by the number of times you want to replicate the recipe: $2\frac{1}{3} \times 2\frac{1}{2} = \frac{7}{3} \times \frac{5}{2} = \frac{35}{6} = 5\frac{5}{6}$ cups of flour.
- 83. d.** Add the needed lengths together using a common denominator of 4: $11\frac{2}{4} + 8\frac{1}{4} + 8\frac{1}{4} = 27\frac{4}{4}$, which simplifies to 28 inches.
- 84. c.** Add the weights together using a common denominator of 12: $3\frac{3}{12} + 8\frac{6}{12} + 4\frac{8}{12} = 15\frac{17}{12}$, which simplifies to $16\frac{5}{12}$ lb.
- 85. c.** Add all four sides of the garden together to find the perimeter. $4\frac{1}{2} + 4\frac{1}{2} + 3 + 3 = 14\frac{2}{2}$, which simplifies to 15 yards. If you chose **b**, you added only *two* sides of the garden.
- 86. a.** Divide the number of packages of cheese (12) by $\frac{1}{4}$ to find the number of pizzas that can be made. Remember to take the reciprocal of the second number and multiply: $\frac{12}{1} \div \frac{1}{4} = \frac{12}{1} \times \frac{4}{1} = \frac{48}{1}$, which simplifies to 48. If you chose **b**, you multiplied by $\frac{1}{4}$ instead of dividing.

- 87. b.** Multiply the number of yards purchased by the cost per yard. Change the mixed number into an improper fraction: $\frac{7}{2} \times \frac{25}{1} = \frac{175}{2}$, which reduces to $87\frac{1}{2}$ or \$87.50.
- 88. b.** Let x stand for the number of tablespoons needed to make $1\frac{1}{2}$ glasses of iced tea. Set up a proportion of the form $\frac{\text{teaspoons}}{\text{glasses}}: \frac{10}{6} = \frac{x}{1\frac{1}{2}}$. Cross multiply and simplify: $10(1\frac{1}{2}) = 6x$; $10 \times \frac{3}{2} = 6x$; $15 = 6x$. Solve for x by dividing both sides by 6: $x = \frac{15}{6} = 2\frac{1}{2}$ tablespoons.
- 89. d.** Divide the amount of money Lucy made by the number of hours she worked. Change the mixed number to an improper fraction. When dividing fractions, take the reciprocal of the second number and multiply: $195 \div 32\frac{1}{2} = \frac{195}{1} \div \frac{65}{2} = \frac{195}{1} \times \frac{2}{65} = \frac{390}{65}$, which simplifies to \$6.
- 90. a.** To find the area, multiply the length by the width. When multiplying mixed numbers, change the mixed numbers to improper fractions: $5\frac{1}{2} \times 7\frac{1}{2} = \frac{11}{2} \times \frac{15}{2} = \frac{165}{4}$, which simplifies to $41\frac{1}{4}$ sq ft.
- 91. a.** Add the number of hours together using a common denominator of 60: $4\frac{30}{60} + 3\frac{45}{60} + 8\frac{12}{60} + 1\frac{20}{60} = 16\frac{107}{60}$, which simplifies to $17\frac{47}{60}$ hours.
- 92. a.** First, multiply 4 by $\frac{1}{2}$ to find the time taken by the four half-minute commercials: $4 \times \frac{1}{2} = 2$. Then, multiply $\frac{1}{4}$ by 3 to find the time taken by the three quarter-minute commercials: $\frac{1}{4} \times 3 = \frac{3}{4}$. Add the two times together to find the total commercial time. Use a common denominator of 4: $\frac{8}{4} + \frac{3}{4} = \frac{11}{4}$, which simplifies to $2\frac{3}{4}$ minutes.
- 93. c.** Let x represent the amount of the original monthly payment. Since refinancing the home loan results in reducing the monthly payment by $\frac{1}{9}$, it follows that the *new* monthly payment $\frac{8}{9}$ of the *original* monthly payment, or $\frac{8}{9}x$. This is equal to \$1,128. Solve for x by dividing both sides by $\frac{8}{9}$: $x = \frac{\$1,128}{\frac{8}{9}} = \$1,128 \div \frac{8}{9} = \$1,128 \times \frac{9}{8} = \$1,269$.
- 94. c.** Multiply the amount of brown sugar needed for one batch ($\frac{3}{4}$) by the number of batches (3): $\frac{3}{4} \times \frac{3}{1} = \frac{9}{4}$, which simplifies to $2\frac{1}{4}$ cups.

- 95. c.** First, we must determine the number of pages *per minute* that six printers can produce when working together. Call this quantity x . To do so, set up a proportion of the form $\frac{\text{printers}}{\text{pages}}: \frac{1\frac{1}{3}}{15} = \frac{6}{x}$. Cross multiply and solve for x : $(1\frac{1}{3})x = 15(6)$; $\frac{4}{3}x = 90$; $x = \frac{3}{4}(90) = 67\frac{1}{2}$ pages. So, in one minute, six printers can produce $67\frac{1}{2}$ pages. Finally, to determine the number of pages produced in 8 minutes, multiply $67\frac{1}{2}$ by 8: $67\frac{1}{2} \times 8 = \frac{135}{2} \times 8 = 540$ pages.
- 96. c.** The prize money is divided into tenths after the first third has been paid out. Find one third of \$2,400 by dividing \$2,400 by 3; \$800 is paid to the first winner, leaving \$1,600 for the next ten winners to split evenly ($\$2,400 - \$800 = \$1,600$). Divide \$1,600 by 10 to find how much each of the 10 winners will receive: $\$1,600 \div 10 = \160 . Each of the last 10 winners will receive \$160.
- 97. a.** The entire company is one whole or $\frac{8}{8}$. Subtract $\frac{3}{8}$ from the whole to find the fraction of the company that does not work in accounting: $\frac{8}{8} - \frac{3}{8} = \frac{5}{8}$. Recall that you subtract the numerators and leave the denominators the same when subtracting then fractions; $\frac{5}{8}$ is equivalent to $\frac{10}{16}$.
- 98. b.** If Kyle eats 8 cookies, then 36 cookies are left ($44 - 8 = 36$). The part that is left is 36, and the whole is 44. Therefore, the fraction is $\frac{36}{44}$. Both the numerator and denominator are divisible by 4. Divide both parts by 4 to simplify the fraction to $\frac{9}{11}$.
- 99. d.** Find $\frac{1}{8}$ of 12 gallons by multiplying: $\frac{1}{8} \times \frac{12}{1} = \frac{12}{8}$. Recall that 12 is equivalent to 12 over 1. To multiply fractions, multiply the numerators and multiply the denominators. Change $\frac{12}{8}$ to a mixed number; 8 goes into 12 once, so the whole number is 1. Four remains, so $\frac{4}{8}$ or $\frac{1}{2}$ is the fractional part; $1\frac{1}{2}$ gallons are left.
- 100. a.** Compare $\frac{3}{4}$, $\frac{3}{5}$, $\frac{2}{3}$, $\frac{2}{5}$ by finding a common denominator. The common denominator for 3, 4, and 5 is 60. Multiply the numerator and denominator of a fraction by the same number so that the denominator becomes 60. The fractions then become $\frac{45}{60}$, $\frac{36}{60}$, $\frac{40}{60}$, $\frac{24}{60}$. The fraction with the largest numerator is the largest fraction; $\frac{45}{60}$ is the largest fraction. It is equivalent to Joey's fraction of $\frac{3}{4}$.

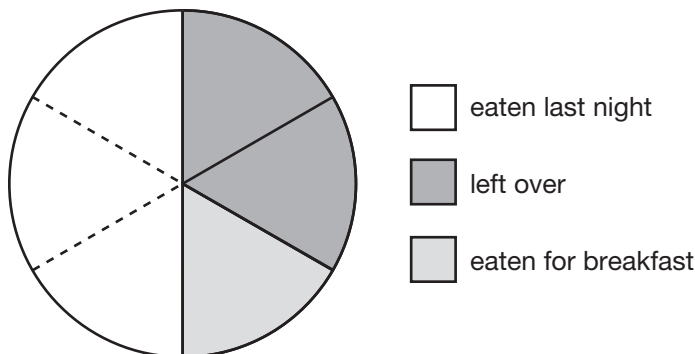
- 101. b.** Break the circle into sixths as shown below; 3 of the sixths are shaded, which is equivalent to $\frac{3}{6}$ or $\frac{1}{2}$.



- 102. d.** To triple the recipe, multiply by 3: $2\frac{3}{4} \times 3 = \frac{11}{4} \times \frac{3}{1} = \frac{33}{4} = 8\frac{1}{4}$. When multiplying a mixed number, change it to an improper fraction first. To find the numerator of the improper fraction, multiply the whole number by the denominator and add the product to the numerator. Keep the denominator the same.
- 103. a.** First, let x represent the number of printers that could, working together, print 1,212 pages in 3 minutes. Such a collection of printers would print 404 pages per minute. Using this, set up a proportion of the form $\frac{\text{printers}}{\text{pages per minutes}}: \frac{1\frac{1}{4}}{20} = \frac{x}{404}$. Cross multiply and simplify: $(1\frac{1}{4}) \times 404 = 20x \times \frac{5}{4} \times 404 = 20x$; $505 = 20x$. Now, solve for x by dividing both sides by 20: $x = \frac{505}{20} = 25\frac{1}{4}$ printers.
- 104. d.** The two pieces of land together are $1\frac{3}{4} + 2\frac{1}{8}$ acres. Add the whole numbers: $1 + 2 = 3$. Add the fractions separately writing each fraction with denominator 8: $\frac{3}{4} + \frac{1}{8} = \frac{6}{8} + \frac{1}{8} = \frac{7}{8}$. Add this to the whole number to get $3\frac{7}{8}$ acres.
- 105. c.** First, since there are 12 months in one year, the number of months are in $4\frac{2}{3}$ years is obtained by multiplying $4\frac{2}{3} \times 12: \frac{14}{3} \times 12 = 56$ months. Next, note that 56 months is comprised of seven 8-month periods, which is found by dividing 56 by 8. However, remember that Allen gets 3 oil changes in each 8-month period, so multiply 7 times 3 to conclude that there are 21 oil changes in this time frame.

- 106. c.** Every whole has 8 eighths. Since there are 8 eighths in each of the 4 wholes, there are 32 eighths in the whole number portion. There are 5 eighths in the fraction portion. Adding the two parts together, you get 37 eighths ($32 + 5 = 37$).
- 107. c.** To double the recipe, multiply the ingredients by 2: $\frac{2}{3} \times 2 = \frac{2}{3} \times \frac{2}{1} = \frac{4}{3}$. Recall that 2 can be written as $\frac{2}{1}$; $\frac{4}{3}$ is an improper fraction. To change it to a mixed number, determine how many times 3 goes into 4. It goes in one time, so the whole number is 1. There is one third left over, so the mixed number is $1\frac{1}{3}$.
- 108. d.** If Mr. Johnson sells $\frac{1}{3}$ of his land, then he still owns $\frac{2}{3}$ of it. To find this, multiply $4\frac{3}{4}$ acres by $\frac{2}{3}$: $4\frac{3}{4} \times \frac{2}{3} = \frac{19}{4} \times \frac{2}{3} = \frac{19}{6} = 3\frac{1}{6}$ acres. In the second step of the multiplication, $4\frac{3}{4}$ was changed to an improper fraction, $\frac{19}{4}$. And in the last step, the improper fraction $\frac{19}{6}$ was converted to a mixed number.
- 109. b.** There are 144 cookies in 12 dozen. First, determine the number of cookies Tim ate by computing $\frac{1}{9}$ of 144; multiply them together to get $\frac{1}{9} \times 144 = 16$. This leaves $144 - 16 = 128$ cookies. Next, Bob takes $\frac{3}{16}$ of the remaining 128 cookies; this equals $\frac{3}{16} \times 128 = 24$ cookies. This leaves $128 - 24 = 104$ cookies. Finally, Fred takes $\frac{4}{13}$ of the remaining 104 cookies; this equals $\frac{4}{13} \times 104 = 32$ cookies. This leaves $104 - 32 = 72$ cookies. Thus, the fraction of the cookies that now remain is $\frac{72}{144} = \frac{1}{2}$.
- 110. b.** The part of their goal that they have raised is \$2,275, and the whole goal is \$3,500. The fraction for this is $\frac{2,275}{3,500}$. The numerator and denominator can both be divided by 175 to get a simplified fraction of $\frac{13}{20}$. They have completed $\frac{13}{20}$ of their goal, which means that they have $\frac{7}{20}$ left to go ($\frac{20}{20} - \frac{13}{20} = \frac{7}{20}$).

- 111. c.** Refer to the drawing below. If half is broken into thirds, each third is one-sixth of the whole. Therefore, she has $\frac{2}{6}$ or $\frac{1}{3}$ of the pizza left over.



- 112. d.** The height of the ball after its first bounce is $\frac{2}{3} \times 108$ feet. The height of the ball after its second bounce is $\frac{2}{3}$ of the previous height, or $\frac{2}{3} \times (\frac{2}{3} \times 108)$ feet. Likewise, the height of the ball after the third bounce is $\frac{2}{3} \times (\frac{2}{3} \times (\frac{2}{3} \times 108))$ feet and after the fourth bounce is $\frac{2}{3} \times (\frac{2}{3} \times (\frac{2}{3} \times (\frac{2}{3} \times 108)))$ feet. Simplifying this last expression yields:

$$\frac{2}{3} \times \left(\frac{2}{3} \times \left(\frac{2}{3} \times \left(\frac{2}{3} \times 108 \right) \right) \right) \text{ feet} = \frac{2 \times 2 \times 2 \times 2}{3 \times 3 \times 3 \times 3} \times 108 \text{ feet} = \frac{16}{81} \times 108 \text{ feet} = \frac{64}{3} \text{ feet} = 21 \frac{1}{3} \text{ feet.}$$

- 113. b.** Multiply 120 by $\frac{3}{5}$: $\frac{120}{1} \times \frac{3}{5} = \frac{360}{5} = 72$. 72 is written as a fraction with a denominator of 1. The fraction $\frac{360}{5}$ is simplified by dividing 360 by 5 to get 72 cups.

- 114. b.** Use a proportion comparing gallons used to time. The plane uses $9\frac{1}{2}$ gallons in 1 hour and the problem asks how many hours it will take to use $6\frac{1}{3}$ gallons.

$$\frac{9\frac{1}{2} \text{ g}}{1 \text{ hr}} = \frac{6\frac{1}{3} \text{ g}}{x \text{ hrs}}$$

To solve the proportion for x , cross multiply, set the cross-products equal to each other and solve as shown below.

$$(9\frac{1}{2})x = (6\frac{1}{3})(1)$$

$$\frac{(9\frac{1}{2})x}{9\frac{1}{2}} = \frac{6\frac{1}{3}}{9\frac{1}{2}}$$

To divide the mixed numbers, change them into improper fractions.

$$x = \frac{\frac{19}{3}}{\frac{19}{2}}$$

$$x = \frac{19}{3} \div \frac{19}{2}$$

$$x = \frac{19}{3} \times \frac{2}{19}$$

The 19s cancel. Multiply the numerators straight across and do the same for the denominators. The final answer is $\frac{2}{3}$ hour.

- 115. a.** There are 12 inches in a foot; $\frac{3}{4}$ of a foot is 9 inches ($\frac{3}{4} \times 12 = 9$). The length of the room is 12 feet 9 inches.
- 116. c.** Multiply \$15,300,000 by $\frac{2}{3}$: $\frac{\$15,300,000}{1} \times \frac{2}{3} = \frac{\$30,600,000}{3} = \$10,200,000$
- 117. b.** Every $\frac{1}{4}$ inch on the map represents 150 miles in real life. There are 4 fourths in every whole ($4 \times 3 = 12$) and 2 fourths in $\frac{1}{2}$ for a total of 14 fourths ($12 + 2 = 14$) in $3\frac{1}{2}$ inches. Every fourth equals 150 miles. Therefore, there are 2,100 miles ($14 \times 150 = 2,100$) between Springfield and Oakwood.
- 118. d.** The width of the opening of the frame is 8 inches. The painting only fills $4\frac{1}{2}$ inches of it. There is an extra $3\frac{1}{2}$ inches ($8 - 4\frac{1}{2} = 3\frac{1}{2}$) to be filled with the mat. There will be an even amount of space on either side of the painting. So, divide the extra space by 2 to find the amount of space on each side because it will be perfectly centered. To divide a mixed number by a whole number, change the mixed number to an improper fraction; $3\frac{1}{2}$ becomes $\frac{7}{2}$. Also, change 2 to a fraction ($\frac{2}{1}$). Then take the reciprocal of 2 and multiply: $\frac{7}{2} \times \frac{1}{2} = \frac{7}{4} = 1\frac{3}{4}$ inches.
- 119. b.** Divide $\frac{1}{2}$ pound by 3. Recall that 3 can be written as $\frac{3}{1}$: $\frac{1}{2} \div \frac{3}{1}$. When dividing fractions, take the reciprocal of the second fraction and multiply: $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$ pound of jellybeans per teammate.
- 120. d.** Multiply $1\frac{1}{2}$ by 10. Change $1\frac{1}{2}$ to an improper fraction ($\frac{3}{2}$) and make 10 into a fraction by placing it over 1 ($\frac{10}{1}$): $\frac{3}{2} \times \frac{10}{1} = \frac{30}{2} = 15$ feet. Each side is 15 feet long, so the dimensions are 15 ft by 15 ft.

- 121. a.** Divide 10 by $\frac{2}{5}$. Recall that 10 can be written as $\frac{10}{1}$: $\frac{10}{1} \div \frac{2}{5}$. When dividing fractions, take the reciprocal of the second fraction and multiply: $\frac{10}{1} \times \frac{5}{2} = \frac{50}{2} = 25$ bows. If you can't figure out what operation to use in this problem, consider what you would do if you had 10 yards of ribbon and each bow took 2 yards. You would divide 10 by 2. This is the same for this problem, except that the bow takes a fraction of a yard.
- 122. a.** Multiply $\frac{3}{4}$ by $\frac{1}{2}$ to find half of the land: $\frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$ acre.
- 123. b.** Subtract Friday's price from Monday's price: $26\frac{3}{8} - 23\frac{1}{16}$. In order to subtract, you need a common denominator. The common denominator is 16. Multiply the first fraction by 2 in the numerator and 2 in the denominator: $\frac{3 \times 2}{8 \times 2} = \frac{6}{16}$. Then subtract: $26\frac{6}{16} - 23\frac{1}{16} = 3\frac{5}{16}$.
- 124. b.** The idea each time is to multiply the fraction by the amount of orange juice in the glass of the child doing the pouring to get the amount in the next child's glass. To begin, the first child starts with 8 ounces of orange juice. The second child in line gets $\frac{3}{4}$ of this amount, which is $\frac{3}{4} \times 8$ ounces = 6 ounces. Next, the third child in line gets $\frac{1}{2}$ of the amount in the second child's glass, which is $\frac{1}{2} \times 6$ ounces = 3 ounces. Finally, the fourth child gets $\frac{2}{3}$ of the amount in the third child's glass, which is $\frac{2}{3} \times 3$ ounces = 2 ounces.
- 125. c.** Multiply the cost per acre by the number of acres: $\$60,000 \times 1\frac{3}{4}$. Change 60,000 to a fraction by putting it over 1 and change the mixed number to an improper fraction: $\frac{\$60,000}{1} \times \frac{7}{4} = \frac{\$420,000}{4} = \$105,000$.